Profiling On-Policy Self-Distillation

General statistics for OPD / OPSD experiments, and statistics for the PMI / prefix-compatibility hypothesis

Outline

Part I: general statistics

0. Setup, notation, cost tags
A. Sanity checks
B. Where the signal lands
C. Estimator quality
D. Gradients and optimizer
E. Calibration and confidence
F. Representations and mechanisms
G. Generalization, forgetting, robustness
H. Experimental rigor

Putting it together

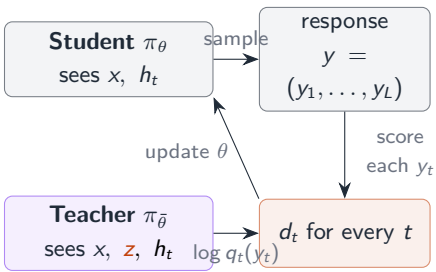
Part II: idea 003 (PMI, compatibility)

II.0 Notation and hypotheses
II.1 Training health and task metrics
II.2 The per-token signal
II.3 Teacher–student distribution gap
II.4 PMI decomposition, prefix compatibility
II.5 Prefix reliability: branch continuations
II.6 Reasoning behavior and diversity
II.7 Parameter dynamics and cost
II.8 Slicing, schedule, staging

Every symbol is defined before it is used; the appendix lists acronyms, symbols and references.

Setup and Notation

Setup: On-Policy Self-Distillation (OPSD)



- x: prompt. z: privileged information (e.g. a reference solution), shown only to the teacher.
- The student π_θ (parameters θ) samples a response $y = (y_1, \ldots, y_L)$; $h_t = (y_1, \ldots, y_{t-1})$ is the prefix before position t.
- Student: $p_t = \pi_\theta(\cdot \mid x, h_t)$.
  Teacher: the same network with parameters $\bar{\theta}$ and extra context z: $q_t = \pi_{\bar{\theta}}(\cdot \mid x, z, h_t)$.
- Per-token signal $d_t = \log q_t(y_t) - \log p_t(y_t)$: positive where the teacher prefers y_t more than the student does.

Plain OPD uses the same formulas with an external teacher model in place of $\pi_{\bar{\theta}}(\cdot \mid x, z, h_t)$. Precise notation follows on the next five slides.

Notation (1/5): prompts, responses, signals

Symbol	Meaning
x	prompt (problem statement)
z	privileged information shown only to the teacher, e.g. a reference solution or the final answer
$\mathcal{V}$; $a \in \mathcal{V}$	vocabulary; a token
G	number of responses sampled per prompt (group size)
$y^{(i)} = (y_1^{(i)}, \ldots, y_{L_i}^{(i)})$, $i = 1, \ldots, G$	the i-th response to x; L_i is its length in tokens
$h_t^{(i)} = (y_1^{(i)}, \ldots, y_{t-1}^{(i)})$	prefix of response i before position t
$p_{i,t} = \pi_\theta(\cdot \mid x, h_t^{(i)})$, $q_{i,t} = \pi_{\bar{\theta}}(\cdot \mid x, z, h_t^{(i)})$	student and teacher next-token distributions
$d_{i,t} = \log q_{i,t}(y_t^{(i)}) - \log p_{i,t}(y_t^{(i)})$	per-token training signal
y, L, h_t, p_t, q_t, d_t	the same objects with i dropped, when only one response is involved

Notation (2/5): parameters and policies

Symbol	Meaning
θ	current student parameters
θ_k; θ_0	parameters after k optimizer steps; initial parameters
$\Delta\theta_k = \theta_k - \theta_{k-1}$	the k-th parameter update
$\bar{\theta}$	teacher parameters: equal to θ, an exponential moving average of θ, or frozen
π_{old}	the policy that generated the current batch; its log-probabilities are recomputed by the training engine
T	sampling temperature
$u_t \in \mathbb{R}^{\lvert\mathcal{V}\rvert}$	student logits at position t, already divided by T, so $p_t = \text{softmax}(u_t)$, i.e. $p_t(a) = \exp(u_t(a)) / \sum_{a'} \exp(u_t(a'))$
$e_a \in \mathbb{R}^{\lvert\mathcal{V}\rvert}$	one-hot vector of token a

Notation (3/5): calculus and probability

Symbol	Meaning
log, exp	natural logarithm and exponential
$\mathbb{R}$	real numbers
$\mathbb{E}$, Pr	expectation, probability
∇_θ, ∇^2_θ	gradient and Hessian with respect to θ
$\mathrm{sg}(\cdot)$	stop-gradient: the value passes through, the gradient does not
$D_{\mathrm{KL}}(p\|q) = \sum_a p(a) \log \frac{p(a)}{q(a)}$	Kullback–Leibler divergence
$\mathbb{I}[\cdot]$; $\mathrm{sign}(\xi) \in \{-1, 0, 1\}$	1 if the condition holds, else 0; sign of a number ξ
$\cos(\mathbf{v}, \mathbf{w}) = \mathbf{v}^\top \mathbf{w} / (\|\mathbf{v}\| \, \|\mathbf{w}\|)$	cosine similarity of two vectors

Notation (4/5): norms and statistics

Symbol	Meaning
$\|\cdot\|$, $\|\cdot\|_F$, $\|\cdot\|_\infty$; tr	Euclidean norm, Frobenius norm, largest absolute entry; matrix trace
$\|\cdot\|$	absolute value of a number; size of a set; length in tokens of a sequence
mean, std, p_{99}	mean, standard deviation, 99th percentile of a list
$\mathrm{AUROC}(s \to c)$	for scores s_j and binary labels c_j: $\Pr(s_j > s_{j'} \mid c_j = 1,\ c_{j'} = 0)$, ties count $\frac{1}{2}$
ρ_{S}	Spearman correlation: Pearson correlation of the ranks

Notation (5/5): evaluation and network internals

Symbol	Meaning
$c_i \in \{0,1\}$; R_i	verifier verdict for $y^{(i)}$ (1 = correct); its scalar reward, e.g. $R_i = c_i$
$\mathrm{ans}(y^{(i)})$; $\mathcal{S}_i \subseteq \{1,\dots,L_i\}$	extracted final answer; positions of the final-answer tokens
$\mathcal{P}$; N	probe set: fixed prompts with fixed sampling seeds; it is re-run every N training steps
$\mathcal{D}_{\mathrm{train}}$, $\mathcal{D}_{\mathrm{held}}$	training prompts; held-out prompts from the same distribution
$\mathrm{acc}(\mathcal{D})$	pass@1 on a prompt set $\mathcal{D}$: mean of c_i over prompts and samples ("avg@16" when 16 samples per prompt)
$\ell = 1,\dots,N_L$; d_{model}	layer index and number of layers; hidden size
$\phi_t^{(\ell)} \in \mathbb{R}^{d_{\mathrm{model}}}$	hidden state at position t after layer ℓ
Φ_ℓ	matrix whose rows are $\phi_t^{(\ell)}$ for a fixed list of probe tokens

The training loss and what it estimates

Per-token loss, in policy-gradient form:

$$\mathcal{L}_t = -\,\mathrm{sg}(d_t)\,\log p_t(y_t)$$

Its expected gradient is the reverse-KL gradient:

$$\mathbb{E}_{y_t\sim p_t}\big[\nabla_\theta \mathcal{L}_t\big] = \nabla_\theta D_{\mathrm{KL}}(p_t\|q_t)$$

Conditions: the prefix h_t is held fixed, and no gradient flows into $\bar{\theta}$.

Proof. Using $\sum_a \nabla_\theta p_t(a) = 0$ and $\nabla_\theta p_t(a) = p_t(a)\nabla_\theta \log p_t(a)$:

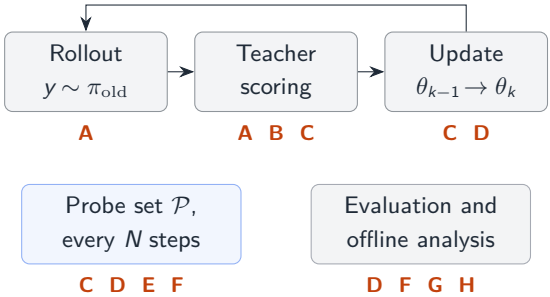
$$\begin{aligned}\nabla_\theta D_{\mathrm{KL}}(p_t\|q_t) &= \sum_{a\in\mathcal{V}} \nabla_\theta p_t(a)\big[\log p_t(a) - \log q_t(a) + 1\big] \\ &= \mathbb{E}_{a\sim p_t}\big[(\log p_t(a) - \log q_t(a))\,\nabla_\theta \log p_t(a)\big].\end{aligned}$$

For the sampled token $a = y_t$ the factor $\log p_t(y_t) - \log q_t(y_t)$ equals $-d_t$, so the term inside the expectation is $-d_t\nabla_\theta \log p_t(y_t) = \nabla_\theta \mathcal{L}_t$.

Section C measures how far this single-sample estimate is from the quantity it estimates.

Cost tags and where each group lives

free	already computed; only needs logging
+fwd	one extra forward pass
logits	probabilities over the vocabulary (or its top tokens)
grad	per-micro-batch gradients or an extra backward
eval	evaluation runs only
offline	on saved checkpoints or dumps
+run	extra training runs
branch	student continuations from forced prefixes



A. Sanity Checks

A. Sanity checks (1/2): is the signal computed correctly?

Statistic	Definition / procedure	Cost	What it tells you
Empty-z regression	Build the teacher input as in training but without z, and score it with $\bar{\theta} = \theta$. Then $q_t = p_t$, so $\max_t \lvert d_t \rvert$ should stay below the noise floor ν (next row).	+fwd once	Wrong concatenation, positions shifted by one, chat-template mismatch.
Noise floor ν	Score the same tokens twice under conditions that should not matter (other batch composition; training vs. inference engine), giving $q_t^{(1)}, q_t^{(2)}$; $\nu = \mathrm{p}_{99} \lvert \log q_t^{(1)}(y_t) - \log q_t^{(2)}(y_t) \rvert$ over scored tokens.	+fwd	Values $\lvert d_t \rvert < \nu$ cannot be told apart from numerical noise.
Token alignment	Fraction of positions whose token id inside the teacher input equals the sampled y_t.	free	Mismatch caused by decoding and re-tokenizing the response.
Teacher-input truncation	Distribution of $\lvert x \rvert + \lvert z \rvert + L$; fraction above the teacher context limit $L_{\max}$; fraction in which part of z is cut.	free	The teacher never saw the full z.

A. Sanity checks (2/2): numerics and bookkeeping

Statistic	Definition / procedure	Cost	What it tells you
Temperature consistency	The student log-probability must use the sampling temperature T (verl divides the logits by T in `dp_actor.py:226`). Choose the teacher's temperature explicitly and log it.	**free**	A mismatch shifts every d_t systematically.
Numerical anomalies	Count of NaN/Inf values; fraction of tokens with $\log p_t(y_t) = 0$ exactly or $\log p_t(y_t) < -30$; $\max_t \|u_t\|_\infty$.	**free**	Overflow and saturation in bf16.
Teacher version	Step at which $\bar{\theta}$ was last copied from θ. For an EMA teacher $\bar{\theta} \leftarrow \beta\bar{\theta} + (1-\beta)\theta$ with decay $\beta \in [0, 1)$, also $\|\bar{\theta} - \theta\|/\|\theta\|$.	**free**	Synchronization works as intended.
Duplicated rows	verl-agent pads a batch by copying rows (`adjust_batch`, `multi_turn_rollout/utils.py:86`); log the copied fraction.	**free**	Copies over-weight some responses in both statistics and gradients.

Line numbers refer to verl-agent at commit 20bd331.

B. Where the Signal Lands

B. Where the signal lands (1/2): concentration

Signal mass of a unit: a token has $|d_{i,t}|$, a response $\sum_t |d_{i,t}|$, a prompt $\sum_{i=1}^{G} \sum_t |d_{i,t}|$. Below, $a_1, \ldots, a_n$ are the masses of the n units at one of these levels.

Statistic	Definition / procedure	Cost	What it tells you
Gini coefficient	$\mathrm{Gini} = \dfrac{\sum_{j=1}^{n} \sum_{j'=1}^{n} \lvert a_j - a_{j'} \rvert}{2n \sum_{j=1}^{n} a_j}$; 0 means perfectly even, close to 1 means one unit holds almost everything. Report all three levels.	**free**	Is the update driven by a handful of tokens, responses or prompts?
Top-1% share	$\sum_{j \in \mathcal{T}} a_j / \sum_j a_j$, where $\mathcal{T}$ holds the 1% of units with the largest a_j.	**free**	The same question, easier to read.
Low-signal prompts	Mean signal per token of a prompt $\bar{a}(x) = \sum_{i,t} \lvert d_{i,t} \rvert / \sum_i L_i$; fraction of prompts with $\bar{a}(x) < \epsilon_{\mathrm{low}}$, a chosen threshold such as $\epsilon_{\mathrm{low}} = \nu$.	**free**	Like all-correct or all-wrong groups in GRPO: full generation cost, almost no gradient.

B. Where the signal lands (2/2): per-prompt dynamics

For a prompt x, k runs over the steps at which x is sampled; $\mathrm{acc}_k(x) = \frac{1}{G}\sum_{i=1}^{G} c_i$ is its fraction of correct responses at step k; $v_k(x) = \mathbb{I}[\mathrm{acc}_k(x) \geq \frac{1}{2}]$ marks it as solved.

Statistic	Definition / procedure	Cost	What it tells you
Per-prompt learning curve	First k with $v_k(x) = 1$; number of forgetting events $\sum_k \mathbb{I}[v_{k-1}(x) = 1,\ v_k(x) = 0]$ (Toneva et al. 2019).	free	Which prompts are learned quickly, and which are learned and then forgotten.
Signal vs. gain	ρ_{S} across prompts between the cumulative signal $\sum_k \bar{a}_k(x)$ ($\bar{a}$ at step k) and the gain $\mathrm{acc}_{\mathrm{last}}(x) - \mathrm{acc}_{\mathrm{first}}(x)$ (last and first sampled step).	free eval	Do prompts that receive more signal improve more?
Effective-token fraction	Tokens that enter the loss after masking, divided by all positions of the padded batch tensor.	free	How much compute produces no gradient.
Train/eval overlap	Fraction of evaluation prompts that share a 13-token span with some training prompt.	offline	Contamination.

C. Estimator Quality

C. Estimator quality (1/2): single sample vs. full distribution

Single-sample estimate (used in training)

$$\hat{g} = \sum_{t=1}^{L} \nabla_\theta \mathcal{L}_t = -\sum_{t=1}^{L} d_t \, \nabla_\theta \log p_t(y_t)$$

Full-distribution gradient (needs q_t on all of $\mathcal{V}$)

$$g^{\text{full}} = \sum_{t=1}^{L} \nabla_\theta D_{\text{KL}}(p_t \| q_t)$$

By the identity on the training-loss slide, $\mathbb{E}[\hat{g} - g^{\text{full}}] = 0$: the gap is pure noise. Both treat every prefix h_t as fixed, i.e. ignore how θ changes later prefixes.

Statistic	Definition / procedure	Cost	What it tells you
Direction agreement	$\cos(\hat{g}, g^{\text{full}})$ per response on $\mathcal{P}$, and again after averaging over the batch. Restrict to one layer's parameters to keep it cheap.	logits grad	How noisy the signal is in gradient space, not just as a scalar.
Within-group gradient spread	$\hat{g}_i$: estimate from response i; $\bar{g} = \frac{1}{G}\sum_i \hat{g}_i$; report $\frac{1}{G}\sum_i \|\hat{g}_i - \bar{g}\|^2 / \|\bar{g}\|^2$.	grad	Is G large enough?

C. Estimator quality (2/2): outcomes and on-policyness

Statistic	Definition / procedure	Cost	What it tells you
Dense signal vs. outcome	$\bar{d}_i = \frac{1}{L_i}\sum_t d_{i,t}$; report AUROC$(\bar{d} \to c)$.	free	Does the per-token judgment agree with final correctness?
Sign agreement with GRPO	$\hat{A}_i = \frac{R_i - \text{mean}(R_1,\ldots,R_G)}{\text{std}(R_1,\ldots,R_G)+10^{-6}}$, $\mathcal{I} = \{i : \hat{A}_i \neq 0\}$, pooled over the batch: $\sum_{i\in\mathcal{I}}\sum_t \mathbb{I}[\text{sign } d_{i,t} = \text{sign } \hat{A}_i] / \sum_{i\in\mathcal{I}} L_i$.	free	With RL mixed in: share of tokens where both signals push the same way.
Off-policy ratio	$r_t = \pi_\theta(y_t \mid x, h_t)/\pi_{\text{old}}(y_t \mid x, h_t)$: its percentiles, and the normalized effective sample size over the n tokens of a mini-batch, $\text{ESS}/n = (\sum_t r_t)^2/(n\sum_t r_t^2) \in (0, 1]$.	free	Drift from on-policy within one step.
Staleness	With asynchronous rollout: number of optimizer steps between the parameters that generated a response and the parameters being updated.	free	Does "on-policy" still hold? (AsyncOPD, f-OPD)

D. Gradients and Optimizer

D. Gradients (1/3): gradient noise scale

g_b: gradient averaged over the b sequences of one accumulation step (all data-parallel ranks); g_B: gradient averaged over all B sequences of the optimizer step.

If per-sequence gradients have mean g and covariance Σ, then $\mathbb{E}\|g_b\|^2 = \|g\|^2 + \mathrm{tr}(\Sigma)/b$, likewise for B; solving both:

$$\widehat{\|g\|^2} = \frac{B\|g_B\|^2 - b\|g_b\|^2}{B-b}, \qquad \widehat{\mathrm{tr}\,\Sigma} = \frac{\|g_b\|^2 - \|g_B\|^2}{1/b - 1/B}, \qquad B_{\mathrm{noise}} = \frac{\widehat{\mathrm{tr}\,\Sigma}}{\widehat{\|g\|^2}}$$

- Smooth numerator and denominator separately over steps before dividing.
- $B \ll B_{\mathrm{noise}}$: a larger batch still cuts the number of steps almost proportionally; $B \gg B_{\mathrm{noise}}$: extra samples per step are mostly wasted (McCandlish et al. 2018).
- Cost **grad**: one extra gradient-norm reduction after the first micro-batch. With token-mean loss aggregation the per-sequence view is approximate.

D. Gradients (2/3): step direction and per-token push

Statistic	Definition / procedure	Cost	What it tells you
Micro-batch consistency	$g_1, \ldots, g_M$: gradients of the M micro-batches of one step, $\bar{g}_M = \frac{1}{M}\sum_m g_m$; report $\frac{1}{M}\sum_m \cos(g_m, \bar{g}_M)$.	grad	Is the step direction stable?
Update persistence	$\cos(\Delta\theta_k, \Delta\theta_{k-1})$, per layer (keeping the previous update of every layer may not fit in memory, so pick a few layers).	free	Oscillating, or moving steadily?
Per-token logit gradient	$\nabla_{u_t}\mathcal{L}_t = -d_t\,(e_{y_t} - p_t)$, so $\lVert\nabla_{u_t}\mathcal{L}_t\rVert = \lvert d_t\rvert\sqrt{1 - 2p_t(y_t) + \sum_a p_t(a)^2}$. Sum it by relative-position bucket t/L and by token type (digits, formula symbols, format tokens, words such as "wait").	logits (the actor already has the logits)	Where parameters are actually pushed: a confident token ($p_t(y_t) \to 1$) gets almost no gradient even when $\lvert d_t\rvert$ is large.

Ignores clipping and the token-aggregation weight of the loss, which only rescale or zero some rows.

D. Gradients (3/3): landscape and plasticity

Statistic	Definition / procedure	Cost	What it tells you
Sharpness	$\lambda_{\max}$: largest eigenvalue of $\nabla^2_\theta \mathcal{L}_\mathcal{P}$, where $\mathcal{L}_\mathcal{P}$ is the sum of $\mathcal{L}_t$ over all tokens of $\mathcal{P}$; power iteration with Hessian–vector products (PyHessian).	offline	Is training moving into sharper regions?
Effective rank of features	$\sigma_1, \ldots, \sigma_r$: singular values of Φ_ℓ; $\bar{\sigma}_j = \sigma_j / \sum_{j'} \sigma_{j'}$; $\mathrm{erank}(\Phi_\ell) = \exp\big(-\sum_j \bar{\sigma}_j \log \bar{\sigma}_j\big)$ (Roy & Vetterli 2007).	+fwd	Collapsing features, a known correlate of lost plasticity.
Weight-norm growth	$\|\theta_k\| / \|\theta_0\|$, per layer.	free	Weight growth also accompanies lost plasticity (Dohare et al. 2024).
Dormant units	Fraction of MLP units (inputs of the down-projection) whose absolute activation stays below a threshold δ on every probe token.	+fwd	Units that stopped contributing.

E. Calibration and Confidence

E. Calibration (1/2): confidence scores and how to grade them

Reverse KL is mode-seeking (Gu et al. 2024), so distilled students tend to become over-confident. Two confidence scores per response:

$$\kappa_i^{\text{tok}} = \exp\Big(\frac{1}{|\mathcal{S}_i|}\sum_{t\in\mathcal{S}_i}\log p_{i,t}\big(y_t^{(i)}\big)\Big), \qquad \kappa_i^{\text{vote}} = \frac{1}{G}\sum_{j=1}^{G}\mathbb{I}\big[\,\text{ans}(y^{(j)}) = \text{ans}(y^{(i)})\big]$$

κ_i^{tok}: geometric-mean probability of the answer tokens. κ_i^{vote}: share of the group with the same final answer.

For n responses with scores κ_i (either kind) and verdicts c_i:

$$\text{ECE} = \sum_{m=1}^{N_{\text{bin}}}\frac{|\mathcal{B}_m|}{n}\Big|\operatorname*{mean}_{i\in\mathcal{B}_m} c_i - \operatorname*{mean}_{i\in\mathcal{B}_m}\kappa_i\Big|$$

$[0,1]$ is split into N_{bin} equal bins; $\mathcal{B}_m$ = responses whose κ_i falls in bin m (Guo et al. 2017).

$$\text{Brier} = \frac{1}{n}\sum_{i=1}^{n}(\kappa_i - c_i)^2$$

Discrimination: AUROC($\kappa \to c$).

All three: **free**

E. Calibration (2/2): over-confidence and uncertainty

Statistic	Definition / procedure	Cost	What it tells you
Wrong-answer confidence	mean$\{\kappa_i : c_i = 0\}$.	free	Is over-confidence getting worse?
Self-certainty	$\frac{1}{L_i}\sum_t D_{\mathrm{KL}}(U \| p_{i,t})$, where U is the uniform distribution on $\mathcal{V}$ (Kang et al. 2025).	logits	Overall confidence level, read from the whole distribution.
Semantic entropy	Cluster the G responses by final answer; a cluster $\mathcal{C}$ has share $\hat{p}_{\mathcal{C}} = \lvert\mathcal{C}\rvert/G$; $\mathrm{SE}(x) = -\sum_{\mathcal{C}} \hat{p}_{\mathcal{C}} \log \hat{p}_{\mathcal{C}}$. Report AUROC$(-\mathrm{SE} \to c^{\mathrm{maj}})$, with $c^{\mathrm{maj}}(x) = 1$ if the largest cluster is correct.	free	Does uncertainty still flag errors? (Simplified from Kuhn et al. 2023, which clusters by entailment.)
Hedging frequency	Occurrences of phrases such as "I'm not sure" or "might be", per 1000 tokens.	free	Has the model stopped expressing doubt?

F. Representations and Mechanisms

F. Representations (1/2): how much, and where, the network changed

Linear CKA (centered kernel alignment) of two matrices X and Y with the same n rows (same tokens in the same order) and centered columns (Kornblith et al. 2019):

$$\mathrm{CKA}(X, Y) = \frac{\|Y^\top X\|_F^2}{\|X^\top X\|_F \, \|Y^\top Y\|_F} \in [0, 1]$$

Statistic	Definition / procedure	Cost	What it tells you
Layer-wise drift	$\mathrm{CKA}(\Phi_\ell$ under θ_0, Φ_ℓ under $\theta_k)$ for every layer ℓ, on the same probe tokens.	+fwd	Are the changes in shallow or deep layers?
Teacher–student alignment	Rows are response positions t. X: student states at layer ℓ given (x, h_t); Y: teacher states at layer ℓ given (x, z, h_t). Plot over training.	+fwd (hook the teacher pass)	Is the student's internal state moving toward what z induces in the teacher?

F. Representations (2/2): mechanisms

Statistic	Definition / procedure	Cost	What it tells you
Attention mass on z	$\mathcal{Z}$: positions of z in the teacher input; $\mathcal{R}$: response positions; $\omega_{t,s}^{(\ell,j)}$: attention weight from position t to s in head j of layer ℓ; N_h heads per layer. $\alpha_\ell = \frac{1}{\lvert\mathcal{R}\rvert N_h} \sum_{t\in\mathcal{R}} \sum_{j=1}^{N_h} \sum_{s\in\mathcal{Z}} \omega_{t,s}^{(\ell,j)}$.	+fwd	How much the teacher reads z; with $\bar{\theta} = \theta$, whether this changes in training.
Logit-lens settling layer	W_U: unembedding matrix; Norm: final normalization. $\ell_t^\star$: smallest ℓ such that $\arg\max_a [W_U \,\mathrm{Norm}(\phi_t^{(\ell')})]_a = \arg\max_a p_t(a)$ for every $\ell' \geq \ell$.	+fwd	Does training move decisions to earlier layers?
Correctness probe	Logistic regression from $\phi_{L_i}^{(\ell)}$ (last response token) to c_i; held-out AUROC for each layer ℓ.	offline	Does the model "know" it is wrong?
Activation magnitude	$\max_{t,\ell} \lVert\phi_t^{(\ell)}\rVert_\infty$ and $\max_t \lVert u_t\rVert_\infty$.	+fwd	bf16 risk (massive activations, Sun et al. 2024).

FlashAttention kernels do not return attention weights, hence eager attention for α_ℓ.

G. Generalization, Forgetting, Robustness

G. Generalization (1/2): what else changed

Statistic	Definition / procedure	Cost	What it tells you
General-capability regression	Before vs. after training on suites such as MMLU-Pro, GPQA, IFEval, HumanEval (or the suites that matter to you).	eval	Were other skills paid for the gain?
Memorization gap	$\mathrm{acc}(\mathcal{D}_{\mathrm{train}}) - \mathrm{acc}(\mathcal{D}_{\mathrm{held}})$.	eval	Learning methods, or memorizing items?
Cross-domain transfer	Train on math; evaluate on code, science, logic.	eval	How far does the gain transfer?
Paraphrase robustness	Write each evaluation prompt in J templates or paraphrases; acc_j: accuracy under template j; report $\mathrm{std}(\mathrm{acc}_1, \ldots, \mathrm{acc}_J)$.	eval	Over-fitted to the training template?

G. Generalization (2/2): harder inputs, facts, imperfect references

Statistic	Definition / procedure	Cost	What it tells you
Difficulty extrapolation	Accuracy on problems harder than the training set (higher difficulty levels, competition sets).	eval	Does the gain extend to harder problems?
Factuality	On SimpleQA-style sets: fraction of attempted answers that are wrong.	eval	Does over-confidence spill over into factual questions?
Noisy-z robustness	Train with a fraction η of the z replaced by wrong references; plot final acc($\mathcal{D}_{\text{held}}$) against η.	+run	Sensitivity to reference quality; real datasets contain wrong solutions.

H. Experimental Rigor

H. Rigor (1/2): uncertainty of the headline numbers

- **Seeds.** At least 3 training seeds per setting; report the mean and a 95% confidence interval (Agarwal et al. 2021).
- **Bootstrap over problems.** Resample problems with replacement, recompute the metric, take the 2.5th and 97.5th percentiles. On a 30-problem set one problem is $1/30 \approx 3.3$ points, so use avg@16 or more.
- **Paired comparison** of methods A and B on the same problems.

$$\chi^2 = \frac{(n_{01} - n_{10})^2}{n_{01} + n_{10}}$$

n_{10}: problems A solves and B does not; n_{01}: the reverse. Compare with the χ^2 distribution with 1 degree of freedom (McNemar 1947).

With avg@16 the per-problem score is not binary; then use a paired bootstrap on per-problem differences instead.

H. Rigor (2/2): noise floors, compute, checkpoint choice

- **Noise floor of probe statistics.** Run $\mathcal{P}$ twice on the same checkpoint with different sampling seeds, giving values $\hat{m}^{(1)}, \hat{m}^{(2)}$ of a statistic. Changes smaller than $|\hat{m}^{(1)} - \hat{m}^{(2)}|$ should not be interpreted.
- **Compute-matched axes.** Besides training steps, plot against cumulative generated tokens, teacher-scored tokens and GPU-hours. OPSD adds a teacher forward pass to every step, so per-step plots flatter it.
- **Fixed checkpoint selection.** Report the final checkpoint, plus the best one chosen by a rule fixed in advance on a validation split. Never select on the test set.

Putting It Together

Schedule: three tiers

Tier	When / data	Statistics
1	every step, full training batch	A: token alignment, truncation, temperature, numerics, teacher version, duplicated rows. B: all except train/eval overlap. C: dense vs. outcome, sign agreement, off-policy ratio, staleness. D: noise scale, micro-batch consistency, update persistence, per-token logit gradient, weight norm. E: confidence scores, ECE, Brier, wrong-answer confidence, semantic entropy, hedging.
2	every N steps, probe set $\mathcal{P}$	C: direction agreement, within-group gradient spread. D: effective rank, dormant units. E: self-certainty. F: drift, teacher–student alignment, attention mass on z, logit lens, activation magnitude.
3	checkpoints and dumps	A: empty-z regression and noise floor (once). B: train/eval overlap. D: sharpness. F: correctness probe. G and H.

Compute tier-1 group statistics on the driver

Per-prompt and per-verdict statistics need the whole batch, but in verl each data-parallel worker sees only a shard.

What to dump at each probe point

Saving raw quantities lets every statistic above be computed offline, and new ones be added later without retraining.

Per token

token id, position t, response index i
$\log p_{i,t}(y_t^{(i)})$, $\log q_{i,t}(y_t^{(i)})$, $\|\nabla_{u_t}\mathcal{L}_t\|$
the 20 most likely tokens and their probabilities under $p_{i,t}$ and under $q_{i,t}$

Per response

c_i, L_i, $\mathrm{ans}(y^{(i)})$, κ_i^{tok}, κ_i^{vote}

Selected layers, sampled positions

$\phi_t^{(\ell)}$ in bf16: $2d_{\mathrm{model}}$ bytes per token per layer.

Size estimate

Two top-20 lists, 4-byte id + 2-byte probability: $2 \times 20 \times 6 = 240$ bytes per token.

Hidden states, $d_{\mathrm{model}} = 1536$ (Qwen2.5-1.5B): about 3 KB per token per layer, so 3 GB per layer for 10^6 tokens. Subsample positions.

Priorities for Part I

1. **All of A.** Almost free, and if any check fails, no other statistic means anything.
2. **From H: several seeds and compute-matched axes.** Reviewers usually ask for both.
3. **G: general-capability regression, and E: wrong-answer confidence.** Forgetting and over-confidence are the two most common side effects of distillation.
4. **The free rows of B, C and E** go into tier 1; **D and F** go into tier 2 on the probe set.

Part II

Statistics for one hypothesis:

privileged-information PMI and prefix compatibility (idea 003)

II.0 Notation and Hypotheses

Part II notation (1/3): decomposing the signal

Teacher network *without* z, with the same prompt template as the student: $q_t^0 = \pi_{\bar{\theta}}(\cdot \mid x, h_t)$.

$$d_t = \underbrace{\log q_t(y_t) - \log q_t^0(y_t)}_{\mathrm{PMI}_t} + \underbrace{\log q_t^0(y_t) - \log p_t(y_t)}_{\Delta_t^{\mathrm{snap}}}$$

- PMI_t (named after pointwise mutual information): how much seeing z raises the teacher's log-probability of y_t.
- Δ_t^{snap} (snapshot gap): the gap between teacher and student parameters.
- If $\bar{\theta}$ is set to θ at rollout time, $\Delta_t^{\mathrm{snap}} = 0$ on the first mini-batch of a step, so $d_t = \mathrm{PMI}_t$ there and PMI_t costs nothing extra. On later mini-batches $\Delta_t^{\mathrm{snap}} = -\log r_t$ (r_t from Section C).

Part II notation (2/3): prefix compatibility

For any policy π, the probability of a prefix is $\pi(h_t \mid x) = \prod_{u=1}^{t-1} \pi(y_u \mid x, h_u)$, and likewise with z in the context. By the chain rule:

$$S_t = \sum_{u=1}^{t-1} \mathrm{PMI}_u = \log \frac{\pi_{\bar{\theta}}(h_t \mid x, z)}{\pi_{\bar{\theta}}(h_t \mid x)}, \qquad S_1 = 0$$

- $S_t > 0$: the prefix is more likely when z is visible; it follows z's route.
- $S_t < 0$: the prefix has left that route, e.g. a different method or an error.
- $S_t \approx 0$: no evidence yet, typical early in the response.

Part II notation (3/3): further symbols

Symbol	Meaning
$\mathcal{H}(p) = -\sum_a p(a) \log p(a)$	entropy of a distribution p over $\mathcal{V}$
$\mathrm{p}_1,\ \mathrm{p}_5,\ \mathrm{p}_{10},\ \mathrm{p}_{50},\ \ldots$	percentiles, defined like p_{99}
Var, Cov, corr	variance, covariance, Pearson correlation over tokens
$T_m(p)$; $\mathrm{rank}_p(a)$	the m most likely tokens; one plus the count of more likely tokens
$\hat{y}_t = \arg\max_a q_t(a)$; eos	teacher's top-1 token at t; end-of-sequence token
$\mathcal{W}$	reflection words: wait, but, hmm, let me check, actually, alternatively
z^+; z^-	correct reference ($z^+ = z$); a wrong one in the same template
$\pi(y \mid x)$, $\pi(y \mid x, z)$	$\prod_{t=1}^{L} \pi(y_t \mid x, h_t)$: whole-response probability (same with z)

Part II: hypotheses to test (none tested yet)

- **H1 (derivability).** OPSD helps when the student could derive z itself (e.g. an image crop). On long reasoning it hurts when the student cannot derive z (e.g. a reference solution), more so the more information z carries.
- **H2 (compatibility, not position).** Whether the teacher's guidance at t helps depends on S_t, not on the relative position t/L.
- **H3 (reflection tokens).** Tokens from $\mathcal{W}$ get negative PMI_t where $S_t \geq 0$, and near-zero or positive PMI_t where $S_t < 0$.

Hypothesis	Key statistics
H1	PMI_t by type of z (II.4); reflection-word rate (II.6); accuracy vs. generation budget (II.1)
H2	reliability AUROC, incremental value, position-stratified AUROC, help vs. compatibility (II.5)
H3	PMI_t of reflection words, split by the sign of S_t and by verdict (II.4)

II.1 Training Health and Task Metrics

II.1 Training health (1/2): is the run stable?

Statistic	Definition / procedure	Cost	What it tells you
Loss and gradient norm	Loss; gradient norm before clipping; fraction of steps skipped because of non-finite gradients.	**free**	Is training stable?
Clip fraction, approximate KL	ϵ_{clip}: PPO clip range. Fraction of tokens with $r_t \notin [1-\epsilon_{\text{clip}}, 1+\epsilon_{\text{clip}}]$; mean of $\log \pi_{\text{old}}(y_t \mid x, h_t) - \log \pi_\theta(y_t \mid x, h_t)$ (verl: `pg_clipfrac`, `ppo_kl`).	**free**	How far one step moves the policy.
Student entropy	$\mathcal{H}(p_t)$: mean, p_{10}, p_{50}, p_{90}, and the fraction of tokens with $\mathcal{H}(p_t) < 0.01$.	**free**	Entropy collapse?

II.1 Training health (2/2): outputs and engines

Statistic	Definition / procedure	Cost	What it tells you
Length and truncation	L: mean, p_{50}, p_{90}, max; fraction of responses cut at the generation limit before eos.	**free**	Does OPSD lengthen or shorten answers? Is the budget enough?
Degenerate output	Fraction of responses with a repeating loop, a format violation (e.g. no `\boxed{}`), or mixed languages.	**free**	Is the output degenerating?
Engine mismatch	π_{inf}: probabilities reported by the inference engine while sampling. Max and p_{99} of $\lvert\log \pi_{\mathrm{inf}}(y_t \mid x, h_t) - \log \pi_{\mathrm{old}}(y_t \mid x, h_t)\rvert$ (`ray_trainer.py:1153`).	**free**	Hidden off-policyness.

II.1 Task metrics (1/2): accuracy and exploration

Statistic	Definition / procedure	Cost	What it tells you
Online accuracy	Mean of c_i on the training batch, logged even when c_i is not part of the loss.	free	Is the student improving?
pass@K curve	K: number of attempts. For a problem with n_{smp} samples, n_{cor} of them correct: pass@$K = 1 - \binom{n_{\mathrm{smp}} - n_{\mathrm{cor}}}{K} \big/ \binom{n_{\mathrm{smp}}}{K}$ (Chen et al. 2021), averaged over problems, for $K = 1, 2, 4, \ldots, 64$.	eval	Does distillation narrow what the student can find?
maj@K	Fraction of problems whose most frequent answer among K samples is correct.	eval	Accuracy with voting.
Exploration gap	pass@64 minus pass@1.	eval	Is diversity being compressed?
Difficulty buckets	$\mathrm{acc}_{\theta_0}(x)$: fraction of correct samples for x under θ_0. Buckets $\{0\}$, $(0, \frac{1}{2}]$, $(\frac{1}{2}, 1)$, $\{1\}$; accuracy per bucket over training.	eval	Which problems gain, which lose?

II.1 Task metrics (2/2): regressions, ceiling, budget

Statistic	Definition / procedure	Cost	What it tells you
Per-problem flips	A problem is solved if at least half of its samples are correct. Between θ_0 and the final checkpoint: counts of unsolved→solved, solved→unsolved, always unsolved, always solved.	eval	Are some problems getting worse?
Imitation gap	acc of the teacher answering with z in context (sampling from $\pi_{\tilde{\theta}}(\cdot \mid x, z)$) minus acc of the student.	eval	How far could the teacher lift the student?
Accuracy vs. budget	acc($\mathcal{D}_{\text{held}}$) at several generation limits.	eval	Does a longer budget hurt? (reported by Rethinking OPSD)
Length by verdict	Mean L_i over responses with $c_i = 1$ and with $c_i = 0$, separately.	free	How length relates to correctness.

Out-of-domain accuracy is covered by Part I, Section G.

II.2 The Signal d_t

II.2 The signal d_t (1/2): distribution

Statistic	Definition / procedure	Cost	What it tells you
Shape	Mean μ_d, standard deviation σ_d, p_1, p_5, p_{50}, p_{95}, p_{99}, min, max, kurtosis $\mathbb{E}[(d_t - \mu_d)^4]/\sigma_d^4$.	**free**	Is the signal heavy-tailed?
Sign and tails	$\Pr(d_t > 0)$, $\Pr(d_t < -5)$, $\Pr(\lvert d_t\rvert > 2)$.	**free**	Mostly pushing tokens up, or down?
Near-zero share	$\Pr(\lvert d_t\rvert < \epsilon_{\mathrm{low}})$, with ϵ_{low} from Section B.	**free**	How many tokens get almost no supervision?
Response-level	$\sum_t d_{i,t}$, $\bar{d}_i$, and the variance of $d_{i,t}$ within a response.	**free**	How much responses differ.

II.2 The signal d_t (2/2): where it falls

Statistic	Definition / procedure	Cost	What it tells you
Position buckets	Split t/L into 10 equal buckets; mean and variance of d_t per bucket.	free	Position effect; PW-OPSD and LGR reach opposite conclusions.
By verdict	Mean $d_{i,t}$ over responses with $c_i = 1$ and with $c_i = 0$.	free	Does the teacher keep rewriting answers that are already correct?
End-of-sequence signal	$\log q_t(\mathrm{eos}) - \log p_t(\mathrm{eos})$ at every position.	logits	Is a length change caused by the teacher pushing to stop earlier or later?
Token categories	Mean d_t for tokens in $\mathcal{W}$, digits, formula symbols, format tokens, and answer tokens ($t \in \mathcal{S}_i$).	free	Which kinds of token carry the signal?

II.3 Teacher–Student Distribution Gap

II.3 Distribution gap (1/2): how far apart are p_t and q_t?

Statistic	Definition / procedure	Cost	What it tells you
Per-token divergences	$D_{\mathrm{KL}}(p_t \Vert q_t)$, $D_{\mathrm{KL}}(q_t \Vert p_t)$; $\mathrm{JSD} = \frac{1}{2} D_{\mathrm{KL}}(p_t \Vert m_t) + \frac{1}{2} D_{\mathrm{KL}}(q_t \Vert m_t)$ with $m_t = \frac{1}{2}(p_t + q_t)$; $\mathrm{TV} = \frac{1}{2} \sum_a \lvert p_t(a) - q_t(a) \rvert$.	logits	Size of the gap; how much switching between forward and reverse KL would matter.
Top-1 agreement	$\Pr\left[\arg\max_a p_t(a) = \hat{y}_t\right]$.	logits	Do both want the same next token?
Teacher's view of the sample	$q_t(y_t)$ and $\mathrm{rank}_{q_t}(y_t)$.	free logits	Does the student often write tokens the teacher finds implausible?
Top-10 overlap	$\lvert T_{10}(p_t) \cap T_{10}(q_t) \rvert / \lvert T_{10}(p_t) \cup T_{10}(q_t) \rvert$.	logits	How similar are the candidate sets?

II.3 Distribution gap (2/2): entropy and approximation error

Statistic	Definition / procedure	Cost	What it tells you
Teacher entropy	$\mathcal{H}(q_t)$, and the 2-D histogram of $(\mathcal{H}(p_t), \mathcal{H}(q_t))$.	logits	Where the teacher is sure and the student hesitates (SEAD partitions tokens this way).
Top-m mass	$\sum_{a \in T_m(q_t)} q_t(a)$ for the m kept by a truncated loss.	logits	Mass ignored by a top-m approximation of the KL.
Single sample vs. exact	$-d_t$ is an unbiased estimate of $D_{\mathrm{KL}}(p_t \Vert q_t)$; report $\mathrm{corr}(-d_t, D_{\mathrm{KL}}(p_t \Vert q_t))$.	logits	Noise of the single-sample signal (vOPD); the gradient-level version is in Section C.

II.4 PMI Decomposition and Prefix Compatibility

II.4 Identity checks: run before anything else

On about 20 responses, before trusting any statistic in this part:

1. **Sum identity.** Check
$$\sum_{t=1}^{L} \mathrm{PMI}_t = \log \pi_{\bar{\theta}}(y \mid x, z) - \log \pi_{\bar{\theta}}(y \mid x)$$
with whole-response log-probabilities computed separately; the error per token should be below 10^{-3}.
2. **Same tokens.** The response token ids are identical in the two prompt templates (with and without z).
3. **Logged signal.** With $\bar{\theta}$ synchronized to θ, the d_t logged during training equals PMI_t recomputed offline.

Cost: **+fwd** once. These complement the general sanity checks of Section A.

II.4 Per-token features (idea 003, F0–F7)

ID	Feature	Definition	Cost
F0	relative position (baseline)	$(t-0.5)/L$	**free**
F1	teacher entropy (baseline)	$\mathcal{H}$ of q_t restricted to $T_{16}(q_t)$, renormalized (PW-OPSD)	**logits**
F2	PMI	PMI_t	**free** / **+fwd**
F3	cumulative PMI	S_t	as F2
F4	mean cumulative PMI	$S_t/(t-1)$ for $t \geq 2$	as F2
F5	local privileged divergence	$D_{\mathrm{KL}}(q_t \| q_t^0)$ on $T_{64}(q_t) \cup T_{64}(q_t^0)$ plus one bucket for the rest	**logits**
F6	candidate PMI	$\log q_t(a) - \log q_t^0(a)$ for $a \in T_3(q_t)$: max and mean	**logits**
F7	contrastive cumulative PMI	$\sum_{u=1}^{t-1} \left[\log \pi_{\bar{\theta}}(y_u \mid x, z^+, h_u) - \log \pi_{\bar{\theta}}(y_u \mid x, z^-, h_u)\right]$	**+fwd**

F2–F4 are **free** when $\bar{\theta} = \theta$. F7 separates the content of z from reference style. II.5 tests F0–F7 as reliability predictors.

II.4 PMI statistics during training (1/2)

Here and on the next slide, **free** applies when $\bar{\theta} = \theta$, **+fwd** otherwise.

Statistic	Definition / procedure	Cost	What it tells you
PMI distribution	Distribution of PMI_t, overall and by position bucket and token category.	**free** **+fwd**	Which tokens does z favor or penalize?
Variance decomposition	$\mathrm{Var}(d_t) =$ $\mathrm{Var}(\mathrm{PMI}_t) + \mathrm{Var}(\Delta_t^{\mathrm{snap}}) + 2\,\mathrm{Cov}(\mathrm{PMI}_t, \Delta_t^{\mathrm{snap}})$.	**+fwd**	With a frozen or EMA teacher, which part drives d_t?
Compatibility profile	Percentiles of S_t against t; fraction of tokens with $S_t < 0$; distribution of S_{L+1} on correct and on wrong responses.	**free** **+fwd**	Are prefixes following z's route?
Compatibility vs. signal	$\mathrm{corr}(S_t, \lvert d_t \rvert)$.	**free** **+fwd**	Is the signal more extreme on incompatible prefixes?

II.4 PMI statistics during training (2/2)

Statistic	Definition / procedure	Cost	What it tells you
Reflection-word PMI (H3)	Mean PMI_t where $y_t \in \mathcal{W}$: correct vs. wrong responses; within wrong ones, $S_t \geq 0$ vs. $S_t < 0$. Control: position-matched tokens not in $\mathcal{W}$.	free +fwd	Does suppression depend on the prefix state?
Contrastive PMI	$\log \pi_{\bar{\theta}}(y_t \mid x, z^+, h_t) - \log \pi_{\bar{\theta}}(y_t \mid x, z^-, h_t)$.	+fwd	Content effect without reference style (RLCSD).
PMI by type of z	Same statistics with $z =$ final answer only, full solution, or a hint.	+fwd per type	More information, more harm? (H1)
Sequence level	With $\bar{\theta} = \theta$: $\mathbb{E}_{y \sim \pi_\theta(\cdot \mid x)}[S_{L+1}] = -D_{\mathrm{KL}}\big(\pi_\theta(\cdot \mid x) \,\|\, \pi_\theta(\cdot \mid x, z)\big)$.	free	Is training pulling $\pi_\theta(\cdot \mid x)$ toward $\pi_\theta(\cdot \mid x, z)$?

II.5 Prefix Reliability: Branch Continuations

II.5 Branch labels: when does following the teacher help?

Spine: a fixed student response (correct: $c = 1$; wrong: $c = 0$). $h \oplus a$: prefix h followed by token a. $\hat{V}(h)$: fraction of n_{roll} student continuations from h, without z, that are correct.

Correct spines (PW-OPSD protocol)

Up to 5 positions per problem with the highest F1; force each $a \in T_3(q_t)$; $n_{\text{roll}} = 6$.
diversity: at least two candidates reach $\hat{V}(h_t \oplus a) \geq 0.75$.
real-uncertain: all candidates, and their mean, below 0.40.
Otherwise **gray**, excluded from AUROC.

Wrong spines (help label)

5 positions per spine, spread evenly over t/L; $n_{\text{roll}} = 8$:

$$\Delta_t = \hat{V}(h_t \oplus \hat{y}_t) - \hat{V}(h_t \oplus y_t)$$

$\Delta_t > 0$: the teacher's top-1 token helps a student without z. If $\hat{y}_t = y_t$, set $\Delta_t = 0$ and report how often this happens.

Cost **branch**: problems $\times 5 \times 3 \times 6$ continuations for correct spines, problems $\times 5 \times 2 \times 8$ for wrong spines.

II.5 Statistics on the branch labels

Statistic	Definition / procedure	Cost	What it tells you
Reliability AUROC	AUROC of each of F0–F7 for real-uncertain vs. diversity, after subtracting each problem's mean score; problem-level bootstrap 95% CI (2000 resamples).	**branch**	Can compatibility beat position?
Incremental value	5-fold logistic regression, folds split by problem: held-out AUROC of F0 plus one feature minus that of F0 alone, with bootstrap CI.	**offline**	Information beyond position.
Position-stratified AUROC	AUROC of F3 and F5 within each third of t/L.	**offline**	Separates compatibility from its correlation with position.
Help vs. compatibility	$\rho_S(\Delta_t, S_t)$; partial correlation given t/L (of residuals after regressing both on t/L); mean Δ_t for $S_t \geq 0$ vs. $S_t < 0$.	**branch**	Does leaving z's route distort guidance (LGR)?
Rescue and endorsement	Rescue: mean $\hat{V}(h_t \oplus \hat{y}_t)$ where $\hat{V}(h_t) \leq 0.1$. Error step: $\hat{V}(h_{t+1}) \leq \hat{V}(h_t) - 0.3$; fraction of error steps with $\mathrm{PMI}_t > 0$.	**branch**	Can the teacher rescue a failing student? Does it endorse mistakes?

II.5 Pre-registered criteria (from idea 003)

Result	Conclusion
AUROC of F3 or F5 $\geq$ that of F0, and the CI of the increment excludes 0	Supports H2: compatibility adds information beyond position
Increment CI includes 0, but F3/F5 AUROC > 0.6 within position thirds	Compatibility largely overlaps with position; H2 holds only partly
F3/F5 AUROC ≈ 0.5 within position thirds	H2 fails; the position effect has another cause
Partial correlation of Δ_t and S_t given t/L clearly positive	Supports "leaving z's route distorts guidance"
Δ_t unrelated to S_t	The LGR decay is not explained by compatibility
Reflection-word PMI clearly higher on $S_t < 0$ segments	Supports H3: masking by prefix state is worth trying
F7 beats F3	Style is a source of noise; the contrastive baseline is needed

Keep positive and negative results. One model on one set of problems is not a general conclusion.

II.6 Reasoning Behavior and Diversity

II.6 Reasoning behavior and diversity (1/2)

Statistic	Definition / procedure	Cost	What it tells you
Reflection-word rate	Occurrences of words from $\mathcal{W}$ per 1000 tokens.	**free**	Fork suppression: does the student stop checking itself?
Self-correction	Fraction of responses whose stated answer changes mid-response; of those, the fraction changed to a correct answer and to a wrong one.	**free**	Is error correction being suppressed?
Same-problem diversity	Among the G responses: number of distinct final answers; self-BLEU (Zhu et al. 2018); mean pairwise embedding distance; mean position of the first token where two responses differ.	**free** / **offline**	Are responses collapsing to one form?

II.6 Reasoning behavior and diversity (2/2)

Statistic	Definition / procedure	Cost	What it tells you
Solution routes	Cluster the G responses by method with an LLM judge; number of clusters.	eval	Is the reference's route the only one left?
Answer leakage	Overlap between y and z: shared n-grams, ROUGE-L, longest common substring; relative position at which the final answer first appears in y.	free	Reasoning or reciting? An answer appearing ever earlier is a warning sign.
Style drift	Perplexity under the initial model, $\exp\big(-\frac{1}{L}\sum_t \log \pi_{\theta_0}(y_t \mid x, h_t)\big)$.	+fwd	How far has the writing moved from the initial model?

Here n in "n-gram" is the span length, e.g. 4. Hedging frequency and semantic entropy are in Section E.

II.7 Parameter Dynamics and Cost

II.7 Parameter dynamics

Statistic	Definition / procedure	Cost	What it tells you
Drift from the initial model	Mean over probe tokens of $D_{\mathrm{KL}}\big(\pi_\theta(\cdot \mid x, h_t) \| \pi_{\theta_0}(\cdot \mid x, h_t)\big)$.	+fwd logits	How far has the model moved?
Teacher drift	$q_t^{(k)}$: teacher distribution with the parameters of step k. On $\mathcal{P}$: $D_{\mathrm{KL}}(q_t^{(k)} \| q_t^{(k-N)})$.	+fwd logits	How fast is the target moving when $\bar{\theta}$ follows θ?
Distillation vs. RL gradients	g_{D}, g_{RL}: gradients of the distillation and RL terms of a mixed loss. $\|g_{\mathrm{D}}\|/\|g_{\mathrm{RL}}\|$, $\cos(g_{\mathrm{D}}, g_{\mathrm{RL}})$, fraction of layers with negative cosine.	grad	Are the two objectives fighting? (Dr.OPD)
Per-layer update size	$\theta^{(\ell)}$: parameters of layer ℓ; $\|\Delta\theta_k^{(\ell)}\|/\|\theta_k^{(\ell)}\|$.	free	Which layers change?
Update structure	For each weight matrix W, its change ΔW over a training window: erank(ΔW) (Section D) and the fraction of entries with $\|\Delta W\|$ below a threshold δ_W.	offline	Is the update low-rank or sparse?

II.7 Cost

Statistic	Definition / procedure	Cost	What it tells you
Token counts	Per step: tokens generated, tokens scored by the teacher, extra tokens scored to obtain q_t^0.	**free**	What each piece costs.
Time	Share of step time spent on teacher scoring; wall-clock per step; GPU-hours to reach a target accuracy.	**free**	Where the time goes.
Truncated supervision	If the loss keeps a subset of tokens and reweights them by inverse inclusion probability (e.g. STOPD): the kept fraction and the largest weight.	**free**	Is the reweighting stable?

II.8 Slicing, Schedule, Staging

II.8 Slicing and schedule

Slice almost every statistic three ways: by verdict c_i; by initial difficulty bucket (II.1); by position bucket of t/L (II.2). Many effects show only inside a slice, e.g. the teacher still restyling responses that are already correct.

Tier	When / data	Part II statistics
1	every step, full training batch	II.1 training health, online accuracy, length by verdict; II.2; II.4 when $\bar{\theta} = \theta$: percentiles of S_t, fraction with $S_t < 0$, reflection-word PMI; II.6 reflection-word rate, self-correction, leakage; II.7 per-layer update size and cost.
2	every N steps, probe set $\mathcal{P}$	II.3; II.4 statistics that need q_t^0 or z^-; II.6 diversity and style drift; II.7 drift and teacher drift.
3	offline	II.1 evaluation metrics; II.4 identity checks; II.5 branch labels and their statistics; II.7 update structure.

II.8 What to add to the dump, and staging

Extra fields (on top of Part I's dump)

Per token: $\log q_t^0(y_t)$, $\mathcal{H}(p_t)$, $\mathcal{H}(q_t)$ as 4-byte floats, plus a 1-byte flag for $y_t \in \mathcal{W}$: 13 bytes.

Per response: type of z; relative position of the first appearance of the final answer.

Per branch: position t, forced token, n_{roll}, $\hat{V}$.

Staging (idea 003)

A. 60 problems: identity checks and correct-spine labels; confirm that the position AUROC is close to PW-OPSD's before comparing new features.

B. Enough problems for at least 40 real-uncertain positions (keep problems whose initial pass rate is in $[0.2, 0.8]$); all branch statistics and H3.

C. Add the tier-1 compatibility statistics to an ongoing OPSD or STOPD run.

Appendix

Acronyms and terms (1/2)

OPD	on-policy distillation: the student samples, a teacher scores every token
OPSD	on-policy self-distillation: the teacher is the student's own network given z
GRPO	group relative policy optimization
RL	reinforcement learning
EMA	exponential moving average
ESS	effective sample size
AUROC	area under the ROC curve
ECE	expected calibration error
CKA	centered kernel alignment
MLP	the feed-forward block of a transformer layer
micro-batch	slice of a batch per backward pass; gradients accumulate over slices

bf16	16-bit floating-point format (bfloat16)
NaN/Inf	not-a-number / infinite floating-point values
FSDP	fully sharded data parallel (training engine)
vLLM	inference engine used for rollouts
verl(-agent)	RL training framework (agent version)
pass@1	probability that one sample is correct
avg@16	pass@1 estimated from 16 samples per problem
FlashAttention	fused attention kernel; unlike plain ("eager") attention it does not return attention weights

Acronyms and terms (2/2)

PMI	pointwise mutual information; here the log-ratio PMI_t of Part II
JSD; TV	Jensen–Shannon divergence; total variation distance
PPO	proximal policy optimization (source of the clip range ϵ_{clip})
LLM	large language model
CI	confidence interval
ROUGE-L	overlap score based on the longest common subsequence
self-BLEU	mean BLEU (n-gram precision) of each response against the other responses of its group
H1–H3; F0–F7	hypotheses (II.0) and per-token features (II.4)
spine; diversity; real-uncertain; gray	branch-label terms, defined in II.5
method names	PW-OPSD, LGR, SEAD, vOPD, RLCSD, Dr.OPD, STOPD, Rethinking OPSD, AsyncOPD, f-OPD: papers in the references

Symbol index (1/3): Part I notation and setup slides

Symbol	Slide
x, z, θ, $\bar{\theta}$	Setup
$\mathcal{V}$, a, G	Notation 1
$y^{(i)}$, L_i, $h_t^{(i)}$	Notation 1
$p_{i,t}$, $q_{i,t}$, $d_{i,t}$	Notation 1
y, L, h_t, p_t, q_t, d_t	Setup, Notation 1
θ, θ_k, θ_0	Notation 2
$\Delta\theta_k$, $\bar{\theta}$	Notation 2
π_θ, $\pi_{\bar{\theta}}$	Setup, Notation 1
π_{old}	Notation 2
T, u_t, e_a	Notation 2
$\mathcal{L}_t$	Training loss

Symbol	Slide
log, exp, $\mathbb{R}$	Notation 3
$\mathbb{E}$, Pr	Notation 3
∇_θ, ∇^2_θ, sg	Notation 3
D_{KL}, $\mathbb{I}$, sign	Notation 3
cos	Notation 3
$\|\cdot\|$, $\|\cdot\|_F$, $\|\cdot\|_\infty$	Notation 4
tr, $\|\cdot\|$	Notation 4
mean, std, p_{99}	Notation 4
AUROC, ρ_{S}	Notation 4

Symbol	Slide
c_i, R_i	Notation 5
ans, $\mathcal{S}_i$	Notation 5
$\mathcal{P}$, N	Notation 5
$\mathcal{D}_{\text{train}}$, $\mathcal{D}_{\text{held}}$	Notation 5
$\text{acc}(\mathcal{D})$	Notation 5
ℓ, N_L, d_{model}	Notation 5
$\phi_t^{(\ell)}$, Φ_ℓ	Notation 5

Symbol index (2/3): Part I, defined where first used

Symbol	Slide
ν, $q_t^{(1)}$, $q_t^{(2)}$	A (1/2)
$L_{\max}$	A (1/2)
β	A (2/2)
a_j, n, Gini, $\mathcal{T}$	B (1/2)
$\bar{a}(x)$, ϵ_{low}	B (1/2)
$\text{acc}_k(x)$, $v_k(x)$	B (2/2)
$\bar{a}_k(x)$	B (2/2)
$\hat{g}$, g^{full}	C (1/2)
$\hat{g}_i$, $\bar{g}$	C (1/2)
$\bar{d}_i$, $\hat{A}_i$, $\mathcal{I}$	C (2/2)
r_t, ESS	C (2/2)

Symbol	Slide
g, Σ, b, B	D (1/3)
g_b, g_B, B_{noise}	D (1/3)
M, g_m, $\bar{g}_M$	D (2/3)
$\lambda_{\max}$, $\mathcal{L}_{\mathcal{P}}$	D (3/3)
σ_j, r, $\bar{\sigma}_j$	D (3/3)
erank, δ	D (3/3)
κ_i^{tok}, κ_i^{vote}	E (1/2)
ECE, N_{bin}, $\mathcal{B}_m$	E (1/2)
Brier	E (1/2)
U, $\mathcal{C}$, $\hat{p}_{\mathcal{C}}$	E (2/2)
SE, c^{maj}	E (2/2)

Symbol	Slide
CKA, X, Y	F (1/2)
$\mathcal{Z}$, $\mathcal{R}$, N_h	F (2/2)
$\omega_{t,s}^{(\ell,j)}$, α_ℓ	F (2/2)
W_U, Norm, $\ell_t^\star$	F (2/2)
J, acc_j	G (1/2)
η	G (2/2)
n_{01}, n_{10}, χ^2	H (1/2)
$\hat{m}^{(1)}$, $\hat{m}^{(2)}$	H (2/2)

Summation indices (j, j', m, s, t, ℓ') are bound by the sum or set in which they appear. The count n is redefined on each slide that uses it (B, C, E, F).

Symbol index (3/3): Part II

Symbol	Slide
q_t^0, PMI_t	II notation 1
Δ_t^{snap}	II notation 1
S_t	II notation 2
$\pi(h_t \mid x)$	II notation 2
$\mathcal{H}$, $\mathrm{p}_1, \mathrm{p}_5, \ldots$	II notation 3
Var, Cov, corr	II notation 3
T_m, rank_p	II notation 3
$\hat{y}_t$, eos, $\mathcal{W}$	II notation 3
z^+, z^-, $\pi(y \mid x)$	II notation 3

Symbol	Slide
ϵ_{clip}, π_{inf}	II.1 health
K, n_{smp}, n_{cor}	II.1 tasks (1/2)
$\mathsf{acc}_{\theta_0}(x)$	II.1 tasks (1/2)
μ_d, σ_d	II.2 (1/2)
m_t, JSD, TV	II.3 (1/2)
F0–F7	II.4 features

Symbol	Slide
$\oplus$, $\hat{V}$, n_{roll}	II.5 labels
Δ_t	II.5 labels
$q_t^{(k)}$	II.7 dynamics
g_{D}, g_{RL}	II.7 dynamics
$\theta^{(\ell)}$, W, ΔW, δ_W	II.7 dynamics

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